

Tutorial: Introduction to Game Theory (1/2)

IEDA 1250 Optimizing Decisions for Personal and Business Development

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June 26, 2026

Today's Scope

Tutorial 1/2

Today is the first game theory tutorial. We focus on intuition, payoff tables, Nash equilibrium, dominated strategies, and pure-strategy zero-sum games.

- What is a game?
- How do we read a payoff matrix?
- What does Nash equilibrium mean?
- How do we simplify games using dominated strategies?
- How do we check for a saddle point in zero-sum games?

Next tutorial

Mixed strategies, graphical method, LP formulation, and Cournot competition can be continued in Game Theory (2/2).

What Makes a Situation a Game?

A game

A decision situation where each player's payoff depends on the actions of more than one decision maker.

Ingredients

- Players
- Strategies
- Payoffs
- Information and timing

Typical assumptions

- Rationality
- Self-interest
- Common knowledge of the game
- Strategic interaction

Why Game Theory?

- In LP and IP, one decision maker optimizes under constraints.
- In game theory, several decision makers optimize at the same time.
- My best decision depends on what you do.
- Your best decision depends on what I do.

Examples

Pricing competition, auctions, political campaigns, traffic routing, and location choices by competing firms.

Example: Prisoner's Dilemma

		Prisoner B	
		Silent	Betray
Prisoner A	Silent	-2, -2	-10, 0
	Betray	0, -10	-5, -5

- First number: payoff to Prisoner A.
- Second number: payoff to Prisoner B.
- Larger payoff is better, so $0 > -2 > -5 > -10$.

Best Response Thinking

Best response

A strategy is a best response if it gives the highest payoff given the other player's strategy.

- If B is silent, A compares -2 with 0 , so A prefers Betray.
- If B betrays, A compares -10 with -5 , so A prefers Betray.
- By symmetry, B also prefers Betray in both cases.

Conclusion

Both players choose Betray.

Nash Equilibrium

Nash equilibrium

A strategy profile where no single player can improve by changing only their own strategy.

- “I am doing my best given what you do.”
- “You are doing your best given what I do.”
- It is a stability concept, not necessarily a fairness or efficiency concept.

Prisoner's Dilemma

The Nash equilibrium is (Betray, Betray).

Equilibrium Can Be Inefficient

		Prisoner B	
		Silent	Betray
Prisoner A	Silent	-2, -2	-10, 0
	Betray	0, -10	-5, -5

- Both staying silent is better for both prisoners than both betraying.
- But if one prisoner is silent, the other wants to betray.
- So (Silent, Silent) is good, but not stable.

Example: Beach Kiosk Game

- A beach is 200 meters long.
- Two vendors choose locations.
- Customers go to the nearest vendor.
- Each vendor wants to attract as many customers as possible.

Question

If the vendors choose locations strategically, where will they locate?

Beach Kiosk Game: Equilibrium Intuition

- If the vendors split the beach evenly at 50m and 150m, each serves half the market.
- But each vendor has an incentive to move closer to the center.
- The stable outcome is that both vendors locate at the center.

Lesson

A Nash equilibrium can be stable for the firms but inconvenient for customers. Stability does not imply social efficiency.

How to Read a Payoff Matrix

		Column player	
		L	R
Row player	U	3, 2	0, 3
	D	2, 1	1, 0

- Row player chooses a row.
- Column player chooses a column.
- First number: row player's payoff.
- Second number: column player's payoff.

Common mistake

Do not compare the two numbers in the same cell. Compare one player's own payoffs while holding the other player's action fixed.

Finding Nash Equilibrium in a 2-by-2 Game

		Player 2	
		L	R
Player 1	U	3, 2	0, 0
	D	1, 1	2, 3

- 1 For each column, mark Player 1's best row.
- 2 For each row, mark Player 2's best column.
- 3 A cell with both players' best responses is a Nash equilibrium.

2-by-2 Game: Solution

		Player 2	
		L	R
Player 1	U	3, 2	0, 0
	D	1, 1	2, 3

- If Player 2 chooses L, Player 1 prefers U.
- If Player 2 chooses R, Player 1 prefers D.
- If Player 1 chooses U, Player 2 prefers L.
- If Player 1 chooses D, Player 2 prefers R.

Result

There are two pure-strategy Nash equilibria: (U, L) and (D, R) .

Dominated Strategies

Dominated strategy

Strategy s is dominated if another strategy is always at least as good and sometimes strictly better, regardless of the opponent's action.

- For a row player who maximizes: larger row payoffs are better.
- For a column player in a zero-sum payoff table for Player 1: smaller entries are better.
- Eliminating dominated strategies can simplify a game before solving it.

Exercise: Eliminate Dominated Strategies

Player 1	Player 2				
	S1	S2	S3	S4	S5
S1	3	2	0	3	0
S2	5	0	2	5	9
S3	7	3	9	5	9
S4	4	6	8	7	5.5
S5	6	0	8	8	3

Task

Assume the entries are payoffs to Player 1. Player 1 maximizes, and Player 2 minimizes. Which rows or columns can we remove?

Exercise: One Elimination Order

- 1 Column S4 is dominated by Column S2.
- 2 Row S1 is dominated by Row S3.
- 3 Row S2 is dominated by Row S3.
- 4 In the reduced table, Row S5 is dominated by Row S3.
- 5 In the reduced table, Column S3 is dominated by Column S5.

Takeaway

Dominance is a simplification method. The order may differ, but each removal must be justified using the remaining table.

Zero-Sum Games

Zero-sum game

One player's gain is exactly the other player's loss.

- A single payoff table is enough: entries are payoffs to Player A.
- Player A wants to maximize the payoff.
- Player B wants to minimize Player A's payoff.
- The value of the game is the expected payoff to Player A under optimal play.

Saddle Point Test

	B1	B2	B3	B4	Row min
A1	20	15	12	35	12
A2	25	14	8	10	8
A3	40	2	10	5	2
A4	-5	4	11	0	-5
Col max	40	15	12	35	

- Player A chooses the row with largest row minimum: $\max \min = 12$.
- Player B chooses the column with smallest column maximum: $\min \max = 12$.
- Since they are equal, there is a saddle point.

Saddle Point Result

Result

The saddle point is $(A1, B3)$, and the value of the game is 12.

- It is the minimum in Row A1.
- It is the maximum in Column B3.
- If A announces A1, B cannot reduce A below 12.
- If B announces B3, A cannot increase above 12.

When There Is No Saddle Point

	B1	B2	B3	Row min
A1	0	-2	2	-2
A2	5	4	-3	-3
A3	2	3	-4	-4
Col max	5	4	2	

- $\max \min = -2$.
- $\min \max = 2$.
- They are not equal, so there is no pure-strategy saddle point.
- This motivates mixed strategies, which can be covered next time.

Today's Checklist

- ① Identify players, strategies, and payoffs.
- ② Read the payoff matrix correctly.
- ③ Use best responses to check Nash equilibrium.
- ④ Remove dominated strategies when possible.
- ⑤ For zero-sum games, try the saddle point test before mixed strategies.

Mini Quiz

- ① In a Nash equilibrium, can the outcome be inefficient?
- ② In a zero-sum game, what does Player B want to do to Player A's payoff?
- ③ What equality must hold for a saddle point?
- ④ Why should we not compare the two payoffs inside the same cell?

Summary

- Game theory studies strategic decision making.
- Nash equilibrium means no player wants to deviate alone.
- Equilibrium can be stable but inefficient.
- Dominated strategies help simplify payoff tables.
- In zero-sum games, a saddle point gives a pure-strategy solution.

Next

Mixed strategies and graphical solution methods are natural next steps when no saddle point exists.